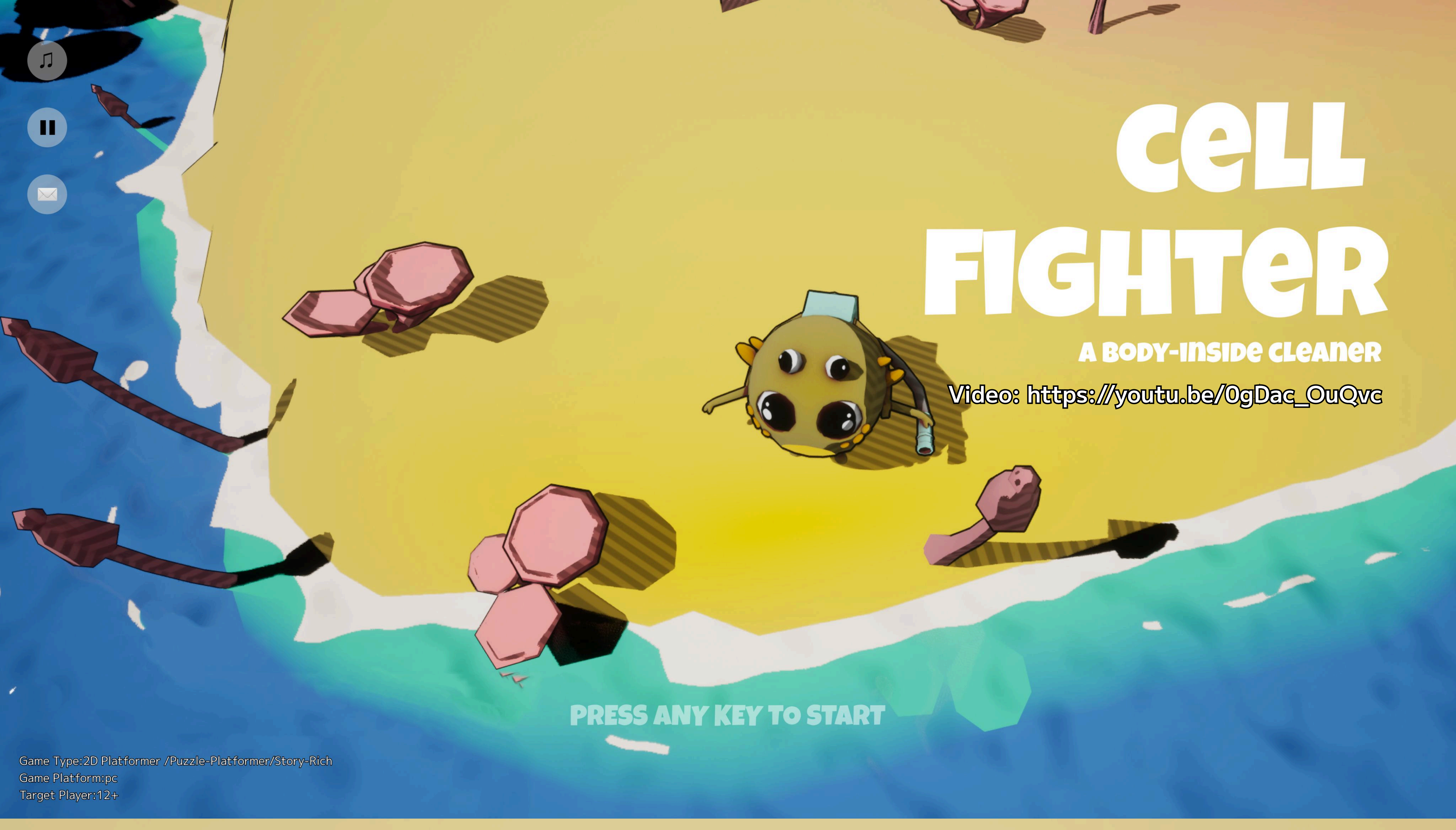

CELL FIGHTER
A BODY-INSIDE CLEANER

Video: https://youtu.be/0gDac_OuQvc

Game Type:2D Platformer /Puzzle-Platformer/Story-Rich
Game Platform:pc
Target Player:12+

INSPIRATION

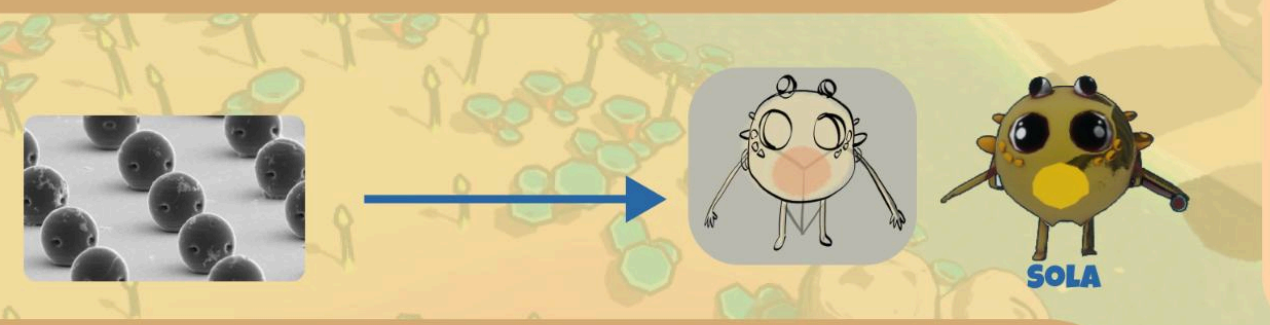
CELLULAR GUARDIANS: THE BODY'S INVISIBLE ARMY

The human body is a complex ecosystem of trillions of cells. Among them, immune cells act as loyal soldiers, patrolling 24/7 to identify and eliminate invading pathogens (like viruses and bacteria) and abnormal internal cells (like cancer cells). For instance, when a virus invades, dendritic cells act as "sentinels," capturing antigens and activating T cells and B cells. B cells produce vast numbers of antibodies, which function like "guided missiles" to neutralize viruses, while T cells become "killer" cells that directly destroy infected host cells. This eternal war raging in the microscopic world is the foundation of our health.

LETHAL THREATS: THE DESTRUCTIVE POWER OF CORONAVIRUSES AND CANCER CELLS

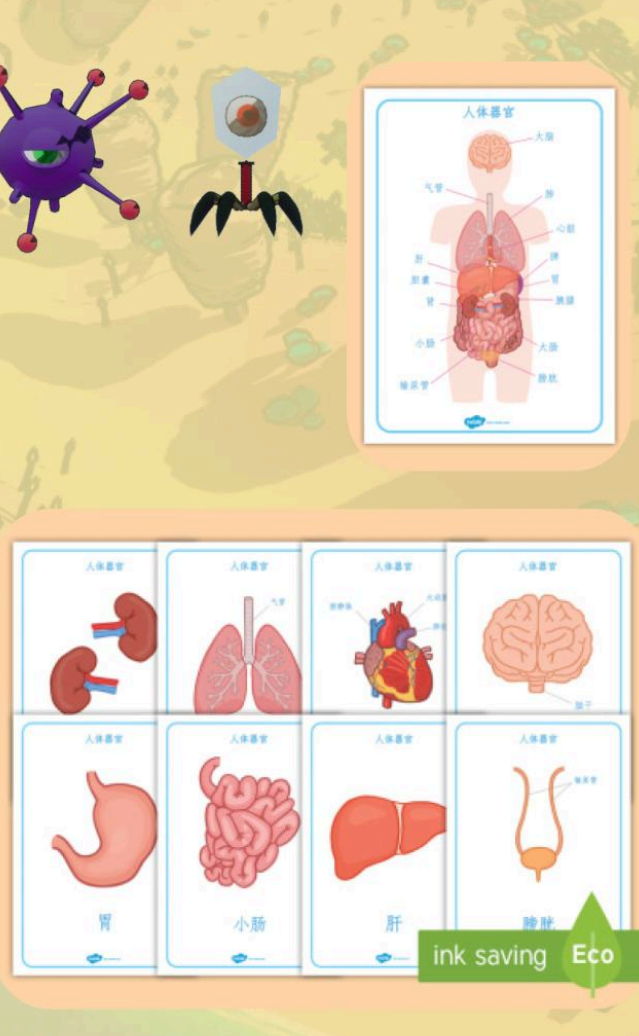
Despite a robust immune system, certain pathogens are exceptionally destructive. Taking the coronavirus as an example, the COVID-19 pandemic has caused over 6 million recorded deaths globally (WHO data), highlighting the global threat of viral respiratory diseases. On the other hand, cancer cells are "traitors" that originate within the body. They form tumors through unlimited proliferation and evading immune surveillance. According to the WHO's International Agency for Research on Cancer (IARC), there were an estimated 20 million new cancer cases and 9.7 million deaths worldwide in 2022. Behind these numbers lies the relentless assault of viruses and cancer cells on the body's intricate systems.

FUTURE HOPE: THE FRONTIER OF MICRO-SCALE THERAPEUTIC ROBOTS



With advancements in nanotechnology, the sci-fi concept of "micro-robots" is inching toward reality. Currently, research teams worldwide are developing micro/nanorobots for targeted drug delivery, minimally invasive surgery, and direct pathogen removal. For instance, scientists have designed magnetic nanorobots that can be guided by external magnetic fields to deliver drugs directly to tumor sites, greatly enhancing efficacy and reducing side effects. In 2023, research demonstrated micro-robots capable of autonomous movement within a hydrogel and cutting cellular debris. While this technology is largely still in the laboratory stage, it represents a revolutionary direction for the future of medicine: a shift from traditional pharmaceutical treatments towards "microscopic mechanical agents" capable of performing repair tasks directly inside our bodies.

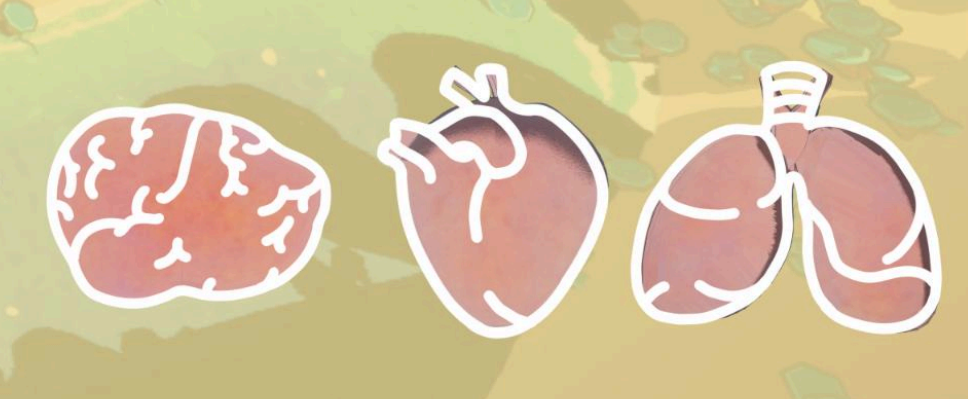
ORGAN VULNERABILITY: THE CRISIS OF DISEASE IN CORE SYSTEMS



When the body's major organs succumb to disease, the entire system of life is endangered.

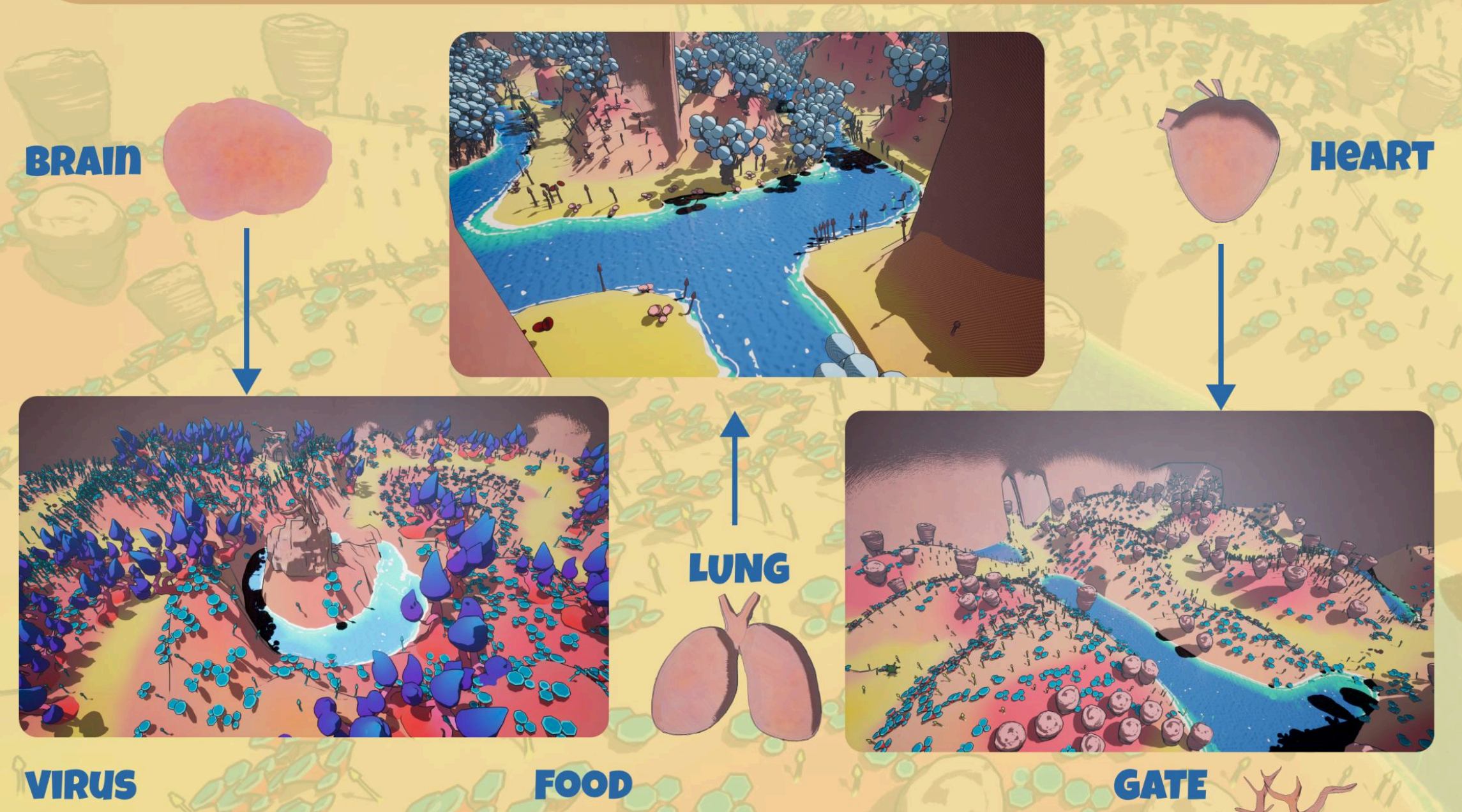
- **The Brain:** It is susceptible not only to organic diseases like stroke and Alzheimer's but is also the seat of mental disorders. An estimated 1 billion people globally live with a mental disorder (WHO), with depression being a leading cause of disability worldwide, demonstrating the complexity of the brain as the "seat of consciousness."
- **The Heart:** Cardiovascular diseases are the world's number one cause of death. Atherosclerosis, which leads to blocked arteries and heart attacks, claims approximately 17.9 million lives each year.
- **The Lungs:** Beyond lung cancer, threats like Chronic Obstructive Pulmonary Disease (COPD) and pneumonia are major concerns. As the primary interface for the body's exchange with the external environment, the lungs are highly vulnerable to infection and pollution.

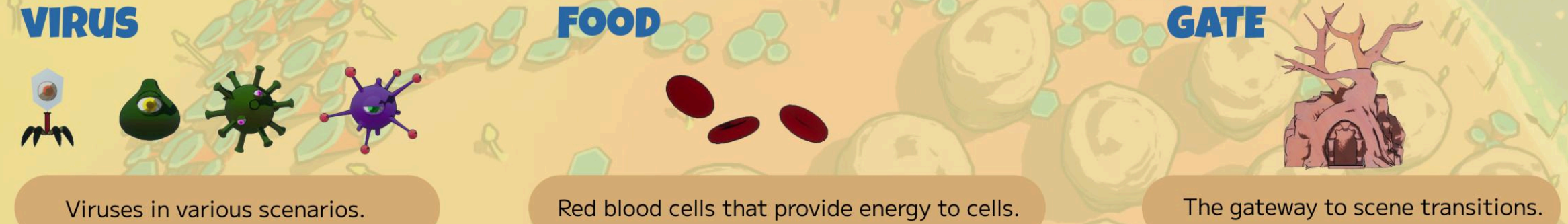
SCENES IN THE SHAPE OF ORGANS



OVERVIEW

"CellFighter" is more than an adventure; it's a love letter to the universe within our own bodies. Through its charming and vibrant cartoon world, we aim to give players of all ages an intuitive sense of the wonder and bustling activity of the human body's microscopic realm. We hope to spark children's curiosity in biological sciences while reminding every player that an epic battle for health is being fought inside them every second—and that they themselves are the ultimate hero of this cosmos. Playing this game is a wonderful journey to understand and appreciate your own body.



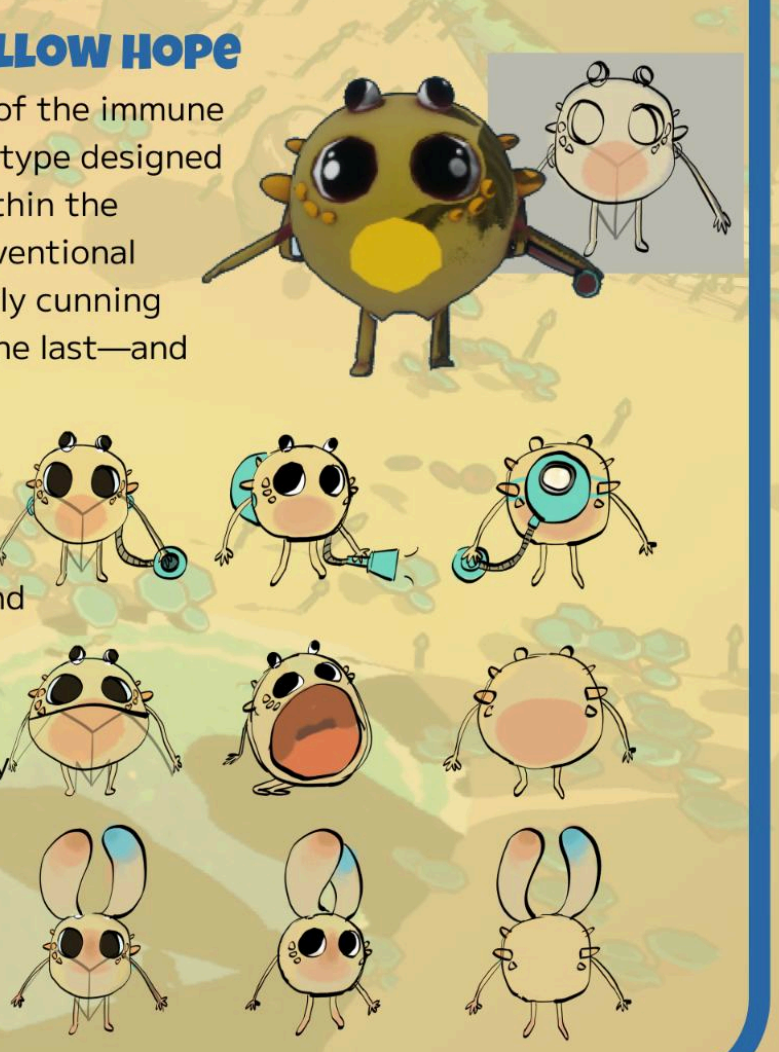


Viruses in various scenarios.

Red blood cells that provide energy to cells.

The gateway to scene transitions.

SOLA
CELLULAR CUSTODIAL AGENT | THE YELLOW HOPE



Sola is not an ordinary cell; she is the embodiment of the immune system's will, a novel micro-therapeutic robot prototype designed for proactive maintenance of the body's health. Within the microscopic universe of the human body, when conventional immune mechanisms are overwhelmed by particularly cunning viruses or severe pathologies, Sola is activated as the last—and smartest—line of defense.

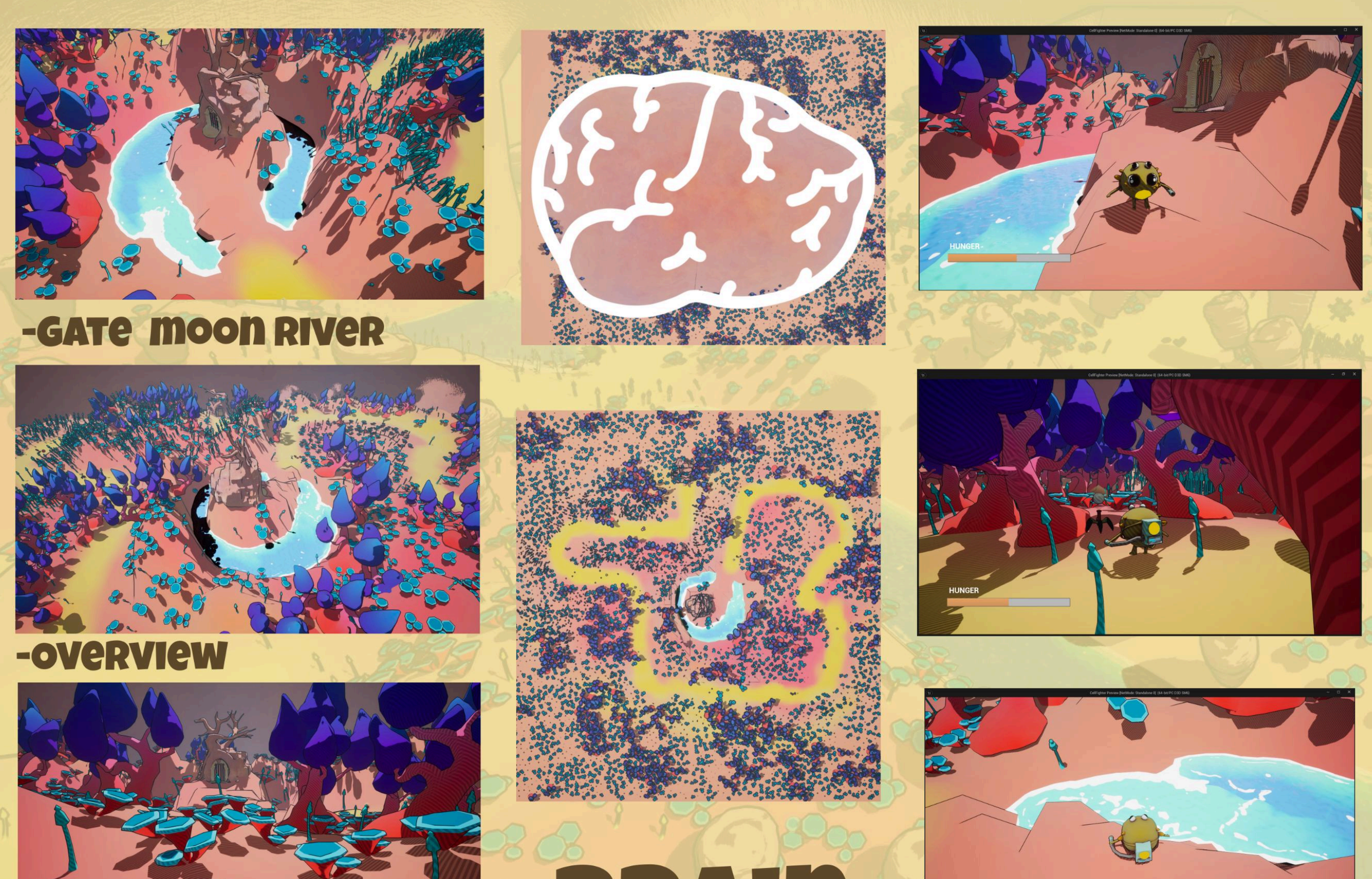
MODE 1: VACUUM MODE
Precise and controlled, this default mode uses the handheld vacuum to efficiently suck away viruses and debris.

MODE 2: GLUTTON MODE
Unleash raw power by opening a giant maw to wildly devour enemies whole, embodying a primal cellular instinct.

MODE 3: MAG-DRAW MODE
Generate a powerful magnetic field to instantly draw in distant viruses for area-of-effect cleansing.

LEVEL1

Welcome to The Brain Dome, the command center of intellect! Here, neon-blue neural grass flickers like thoughts, and strange mushrooms grow like memory nodes. Navigate forests shaped from synapses, and beware the Phage-Spiders lurking in the shadows—they seek to parasitize precious logic and reason. Use your wits to purify this sanctuary of thought.


-GATE MOON RIVER

-OVERVIEW

BRAIN

MAP DETAIL
-VIRUS TYPE

PHAGE-SPIDER
A virus with a transparent polyhedral head supported by a red spine, featuring a single large red eye and multiple black legs. Its design fuses a bacteriophage with a spider.

-ENVIRONMENT ASSETS

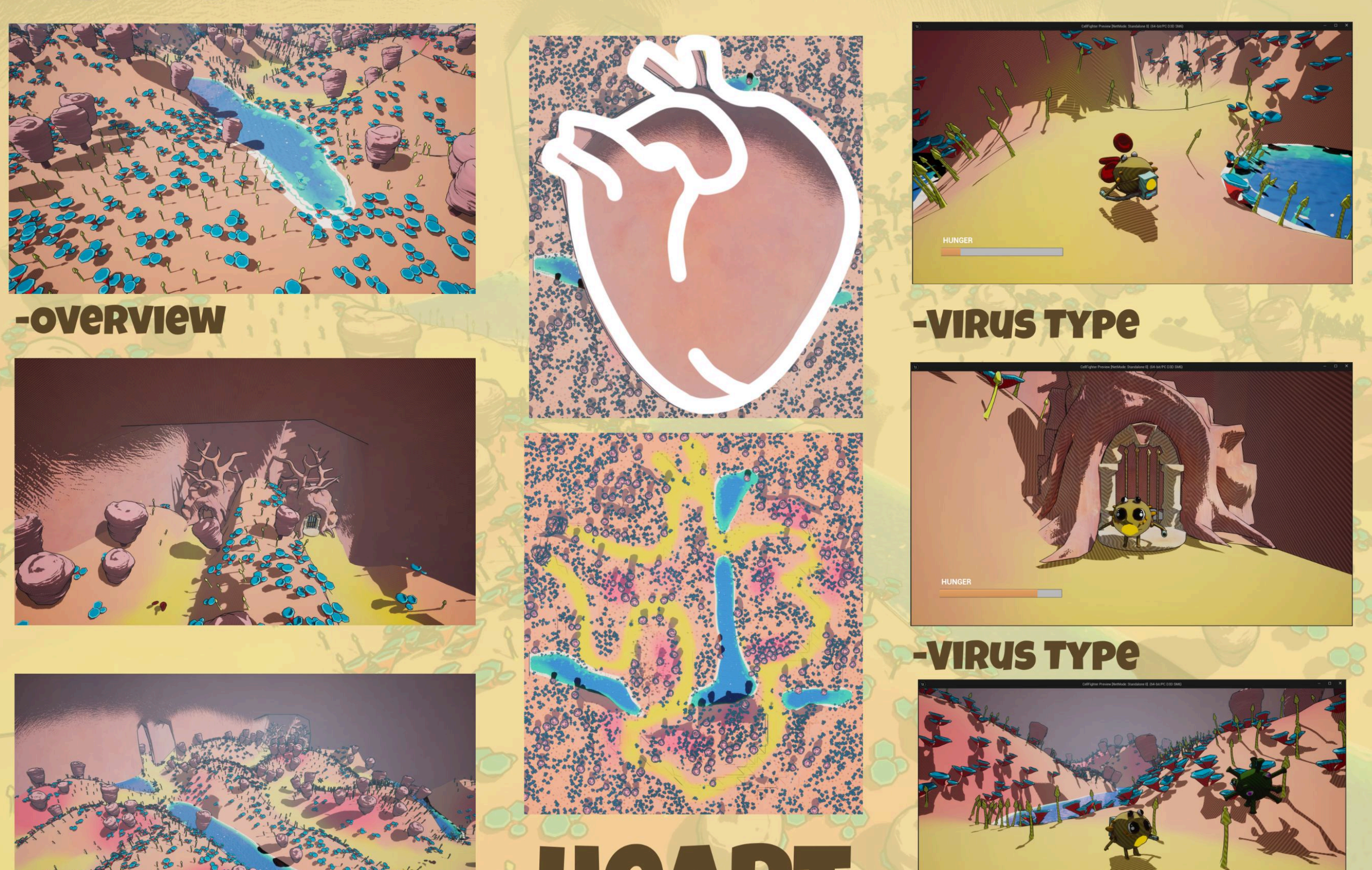
NEURAL-BLUE GRASS
Grass that glows with a faint blue light, shaped like nerve fibers, symbolizing the brain's neurons.

MYCO-MYELIN MUSHROOM
A mushroom with a red stem and blue cap, simulating the insulating function of the myelin sheath.

SYNAPSE PEPPER TREE
A tree with a flesh-colored trunk and a gradient blue-purple crown shaped like a pepper, representing chemical transmission across synapses.

LEVEL2

Welcome to The Brain Dome, the command center of intellect! Here, neon-blue neural grass flickers like thoughts, and strange mushrooms grow like memory nodes. Navigate forests shaped from synapses, and beware the Phage-Spiders lurking in the shadows—they seek to parasitize precious logic and reason. Use your wits to purify this sanctuary of thought.


-OVERVIEW

-VIRUS TYPE

-VIRUS TYPE

HEART

MAP DETAIL
-VIRUS TYPE

CARDIO-DROPLET
A droplet-shaped, green, one-eyed monster, inspired by aggregating pathogens in the blood.

-ENVIRONMENT ASSETS

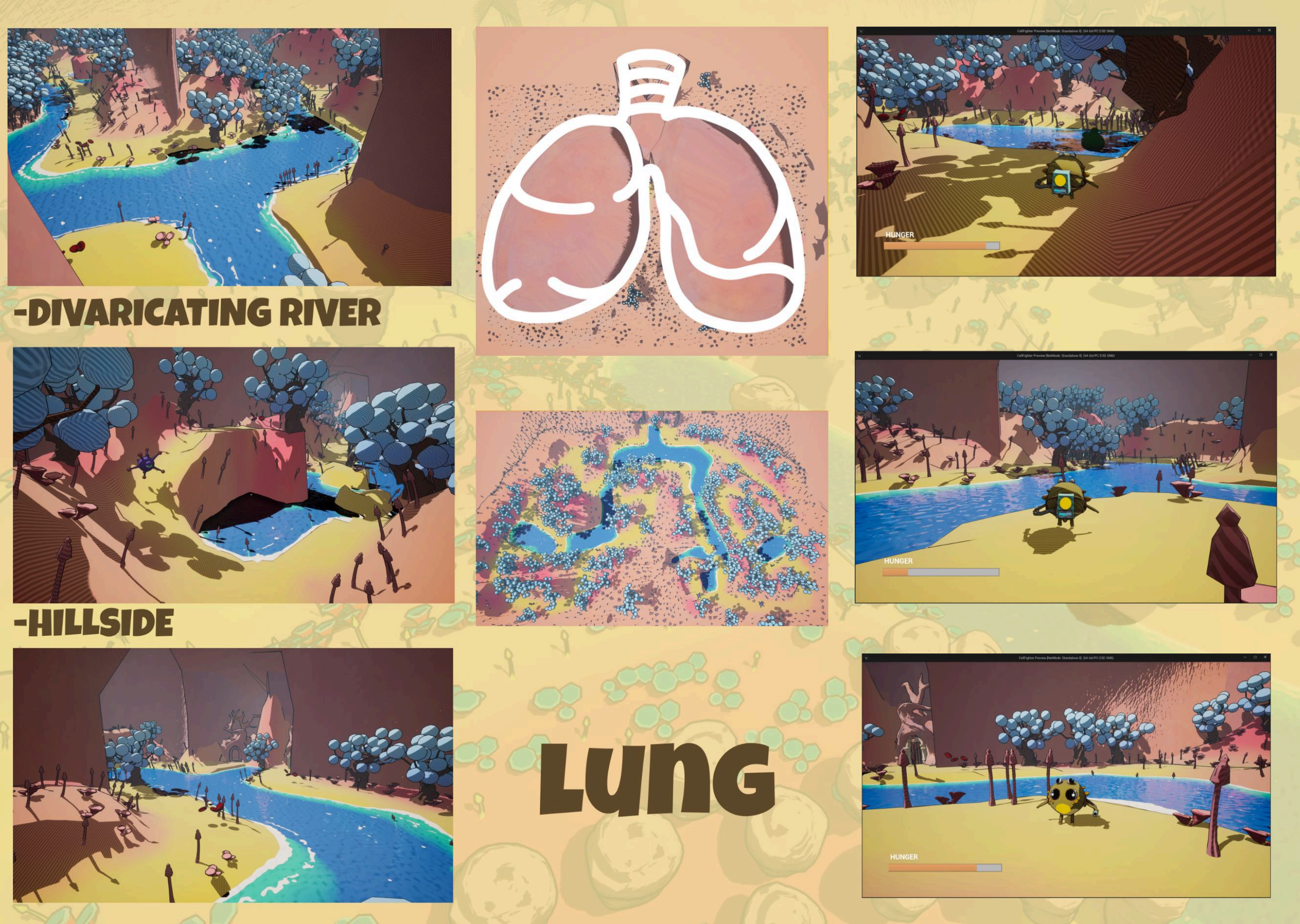
PULSE-YELLOW GRASS
Yellow nerve-like grass that sways to the rhythm of a heartbeat, echoing the heart's electrical signals.

HEMATO-CAP MUSHROOM
The same red-rooted, blue-capped mushroom, here symbolizing the heart's blood sinuses.

MYO-COTTON TREE
A tree with a flesh-colored trunk and a light flesh-colored, cotton candy-like spherical crown, mimicking the fluffy texture of myocardial tissue.

LEVEL3

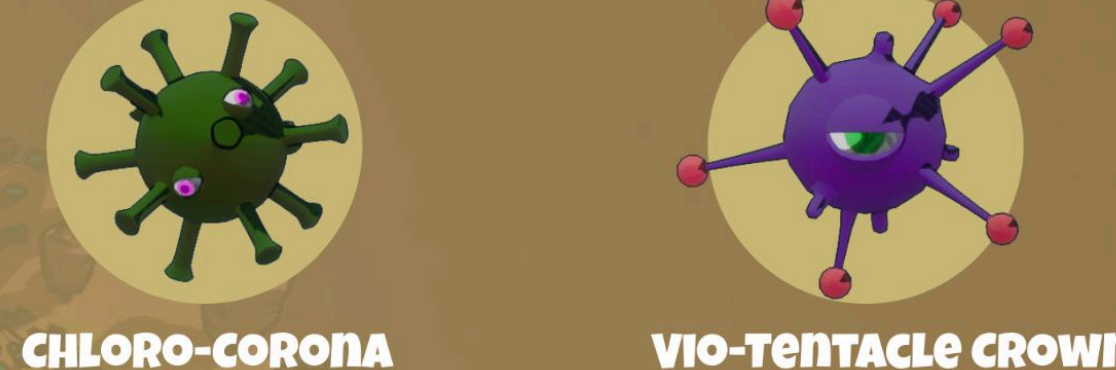
Arrive at last at The Lung Dome, the grand canopy of breath! Flesh-colored grass carpets the ground, and colossal trees bear alveoli-like sacs. Yet, decay lingers here, as the Chloro-Corona and Vio-Tentacle Crown poison the very air of life. Take a deep breath, and use your strength to clear this haze, allowing pure air to flow once more.


-DIVARICATING RIVER

-HILLSIDE

LUNG

MAP DETAIL
-VIRUS TYPE



CHLORO-CORONA
A green, suction-cup covered coronavirus with multiple purple eyes, symbolizing a respiratory adhesive pathogen.

VIO-TENTACLE CROWN
A purple coronavirus with green eyes and pink spherical tentacles on its crown, inspired by mutated envelope proteins.

-ENVIRONMENT ASSETS

ALVEOLI GRASS
Flesh-colored nerve-like grass, simulating the lung's connective tissue.

PNEUMO-MUSHROOM
A flesh-colored Lingzhi mushroom, representing a fungal-like symbiont in the lung cavity.

BRONCHIO-ALVEOLAR TREE
A tree with a brown trunk and branches full of alveoli-like spheres, corresponding to the structure of the bronchioles.

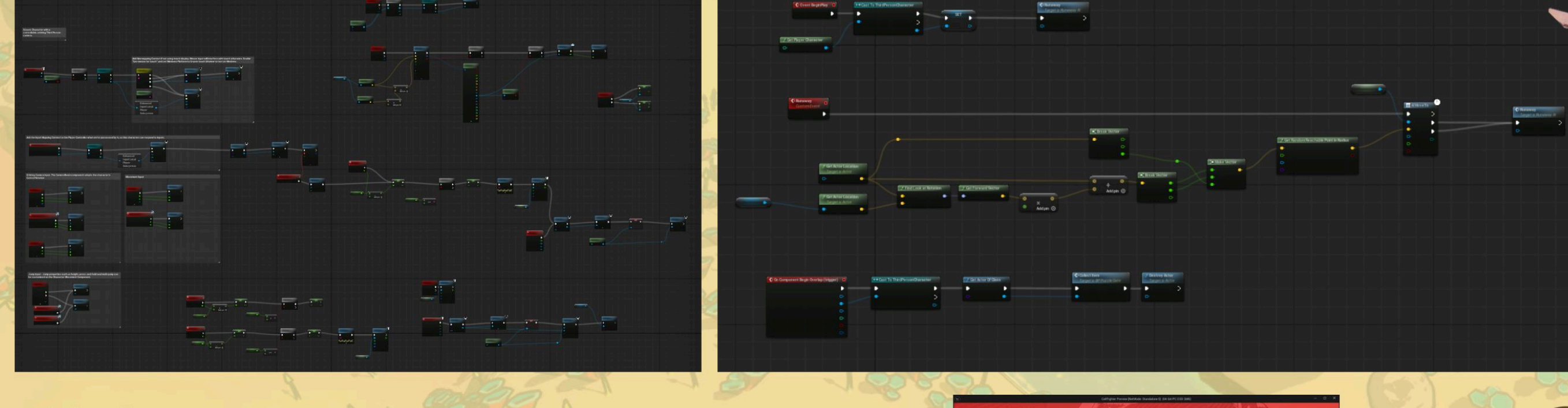
GAME BLUEPRINT

-CHARACTER BLUEPRINT

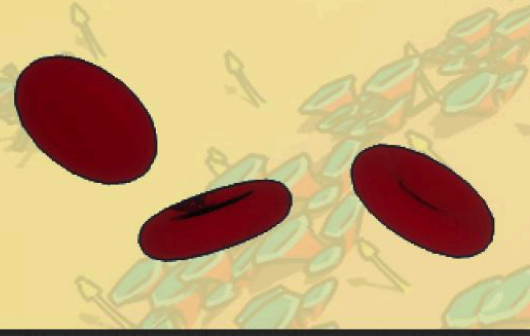
I followed blueprint tutorial videos on YouTube and created my own game blueprints.
I added blueprints for this character to eat food, lose health every second, die when there's no health left, and collect keys to open doors.

-VIRUSES

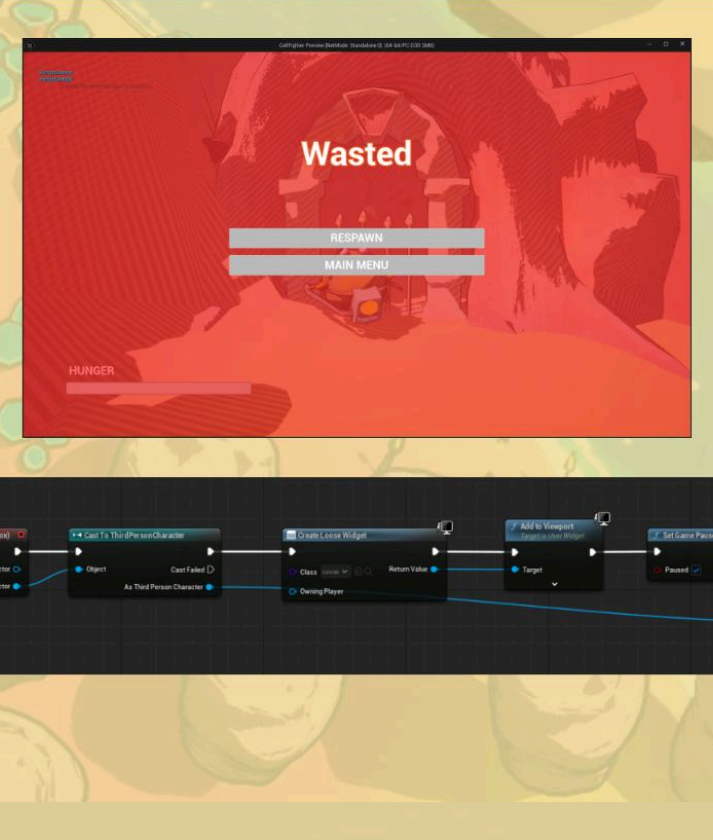
Regarding the virus blueprint, I added a blueprint that allows the player to evade the protagonist's free movement.

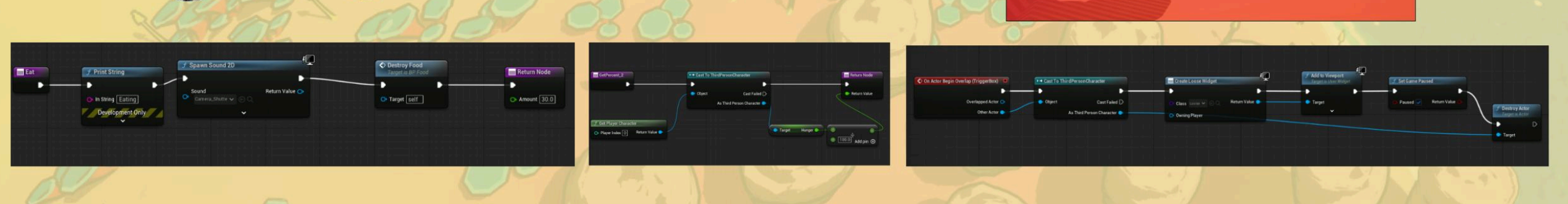


-HEALTH

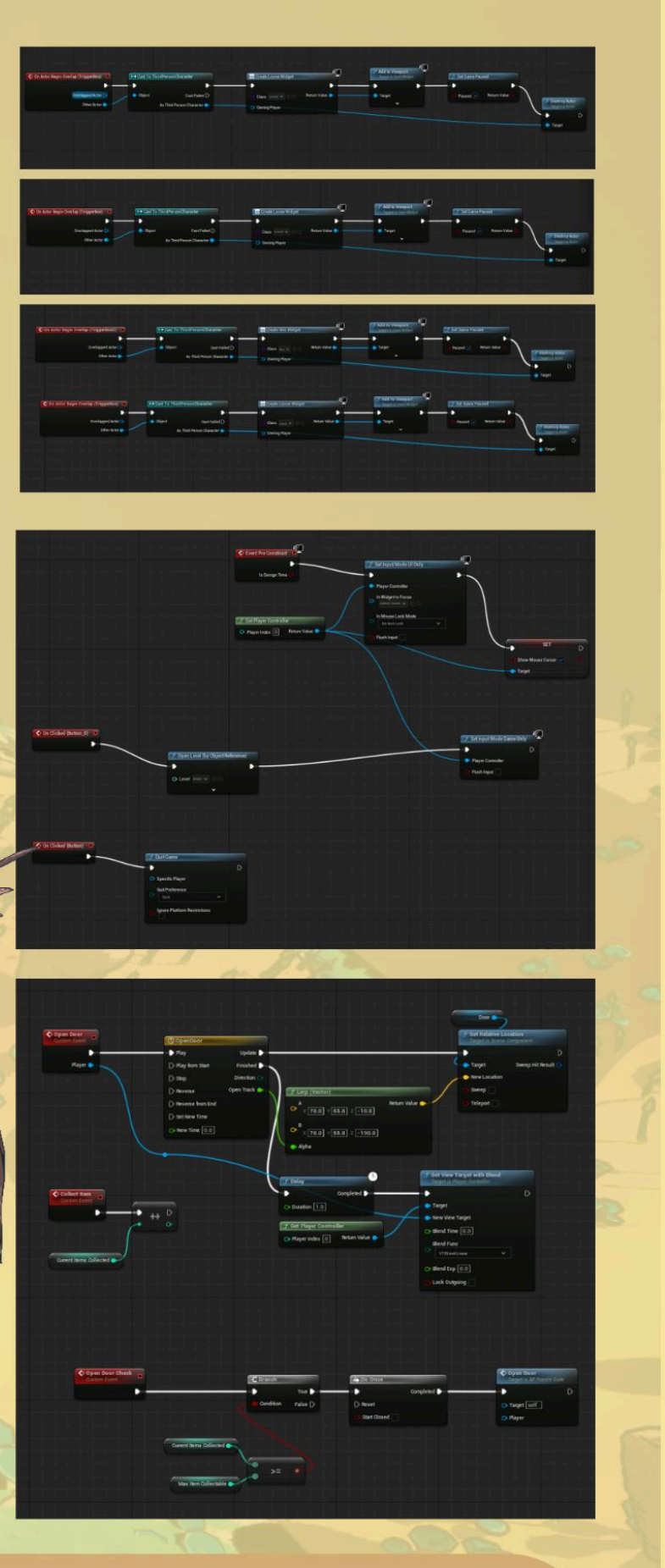


In the protagonist's health blueprint, I added satiety and food. The mesh looks like red blood cells, and the protagonist can restore health by eating them.
If food isn't replenished in time, the protagonist will starve to death, so I've included a failed blueprint here.





-SCENE TRANSITION AND GAME SUCCESS



-PORTAL

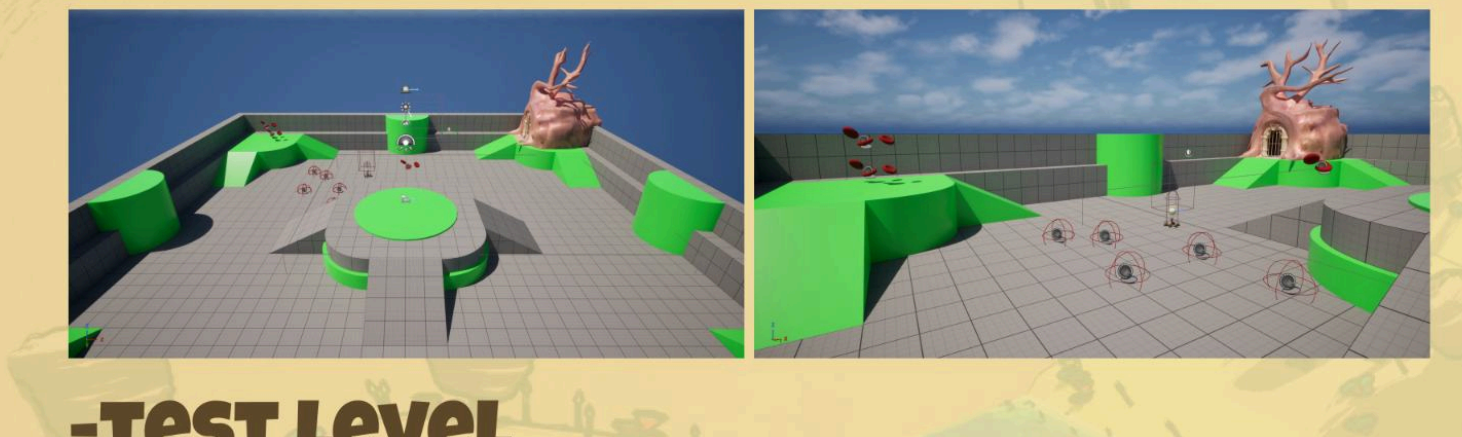
For the scene interactive object blueprint, I created a portal to the next level. A sufficient number of viruses need to be cleared to interact with at the portal before the iron gate will slowly open, thereby enhancing player interactivity and game immersion.

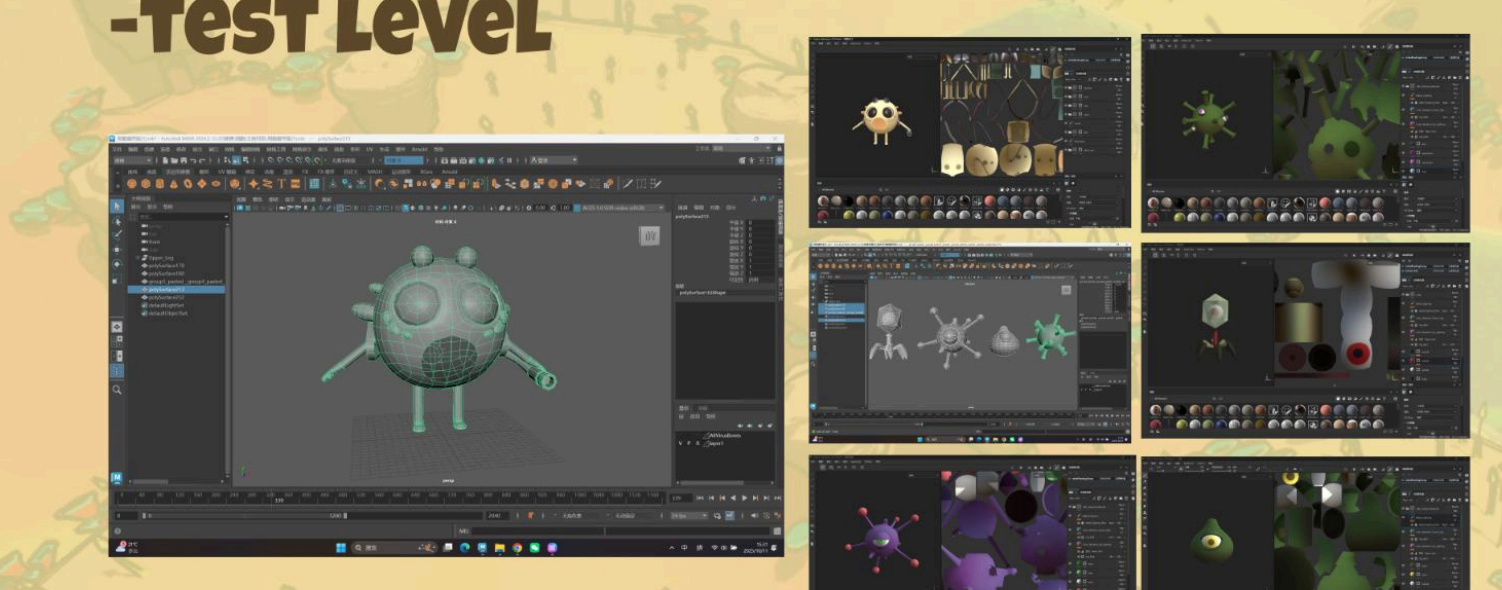
-OTHER ASSETS

WATER SHADER

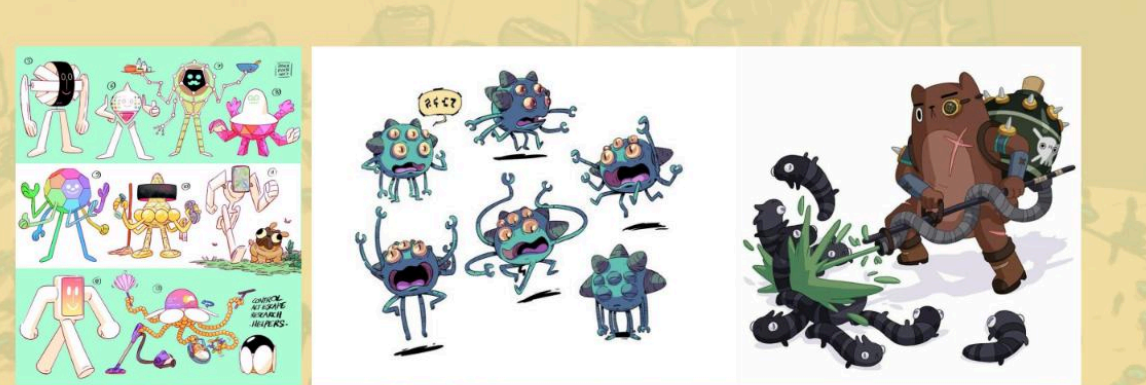
TOON SHADER

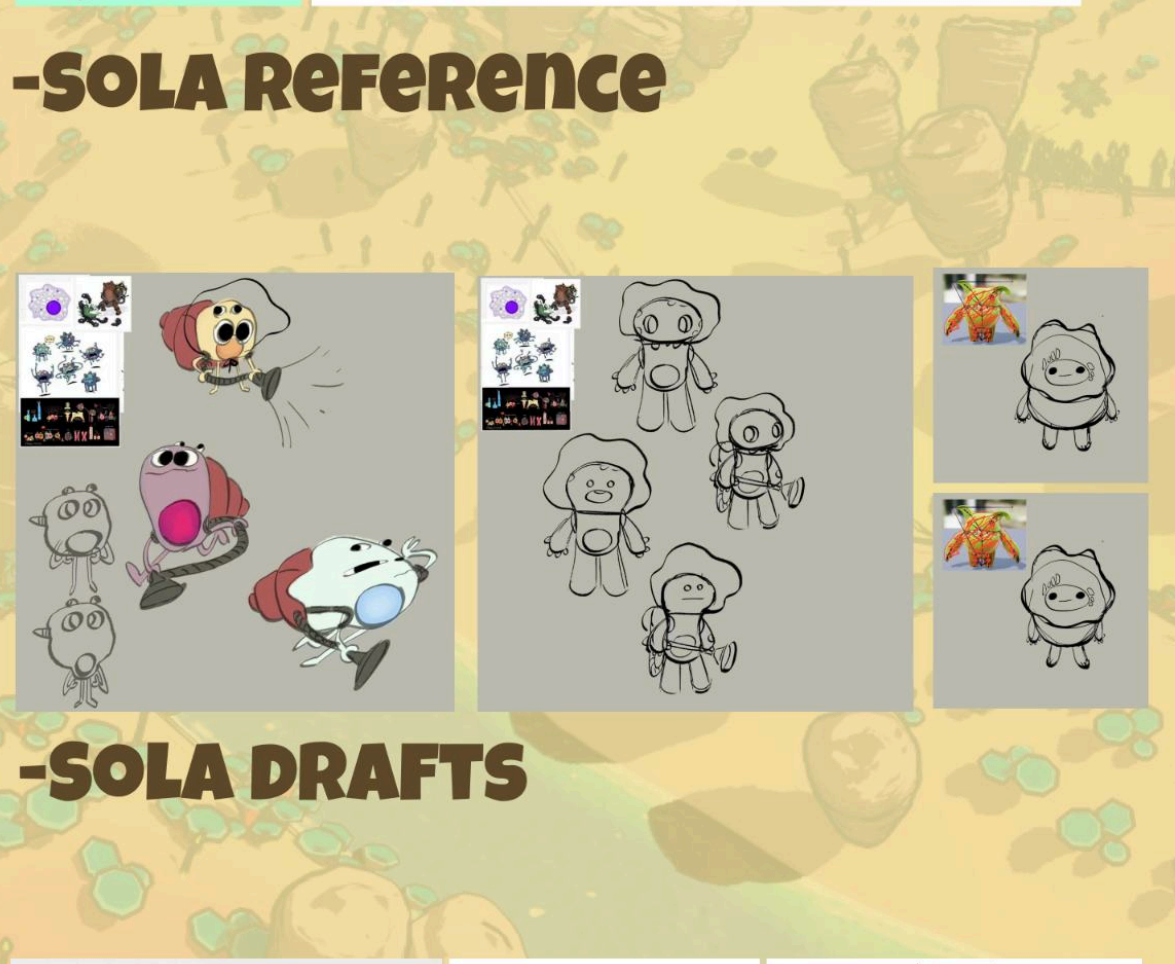
DRAFTS&REFERENCES

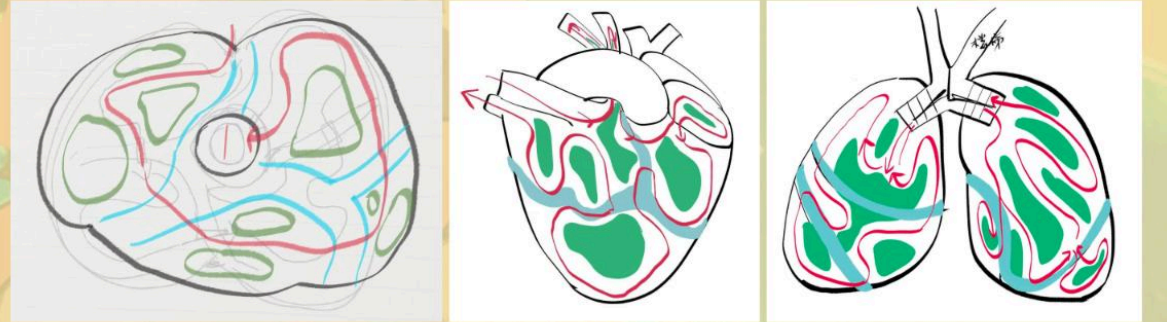

-TEST LEVEL


-MODELING

Das Herz des Menschen als Fabrik
-SCENE REFERENCE


-SOLA REFERENCE


-SOLA DRAFTS


-SOLA DRAFTS

SOMETHING SPOOKY ARRIVING
VERY SOON...
-MAP REFERENCES